

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`
Output: `3`

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`
Output: `2`

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int candidate = 0;
8     int count = 0;
9     int num;
10
11     for (int i = 0; i < n; i++) {
12         scanf("%d", &num);
13
14         if (count == 0) {
15             candidate = num;
16         }
17
18         if (num == candidate) {
19             count++;
20         } else {
21             count--;
22         }
23     }
24
25     printf("%d\n", candidate);
26
27     return 0;
28 }
```

	Input	Expected	Got	
✓	3 3 2 3	3	3	✓

Passed all tests! ✓